Evaluating the Effectiveness of Point-of-Care EEG in Emerging Markets

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Background

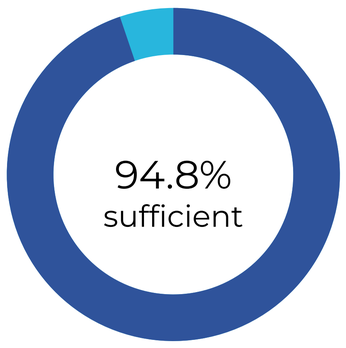
Electroencephalography (EEG) is essential for diagnosing seizures, but access is often limited by high costs and a shortage of specialists. BrainCapture developed the smartphone-based BC-1 EEG system to address this, enabling use by non-expert healthcare workers (NEHWs) with remote interpretation. This study reports on the first large-scale clinical implementation of the BC-1 in Kenyan hospitals.

Materials & Methods

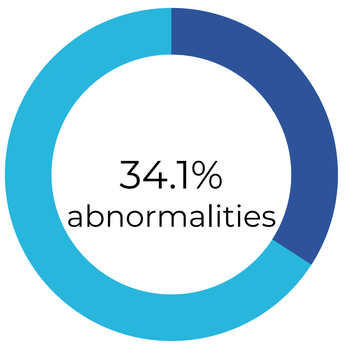
Training was conducted at 20 Kenyan hospitals, where 146 NEHWs were trained to use the BC-1 system. A total of 1,273 EEGs were recorded from 1,239 participants, primarily those with a history of convulsions. Each recording was evaluated by a clinical neurophysiologist for diagnostic quality.



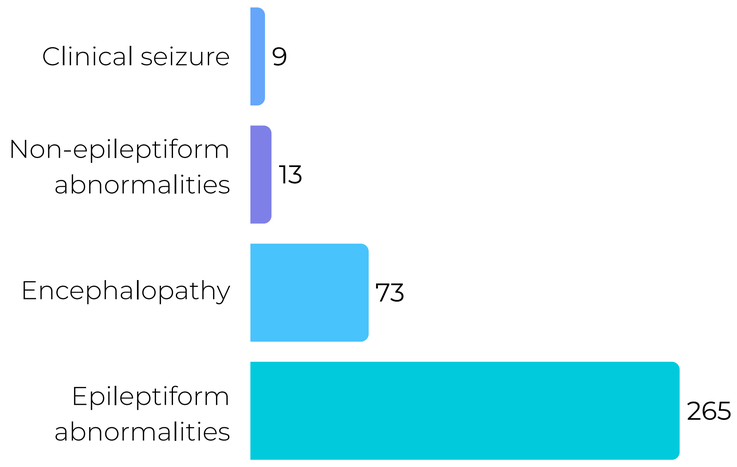




EEG quality



Normal vs abnormal EEG



Breakdown of abnormal EEGs

Results

The BC-1 system enabled the acquisition of high-quality EEGs, with 94.8% (1,194/1,273) of recordings meeting diagnostic quality criteria. Signal noise and movement artifacts were the main causes of insufficient quality.

Abnormalities were detected in 34.1% (407) of the interpretable EEGs, including 265 with epileptiform abnormalities, 73 showing encephalopathy, 9 capturing clinical seizures, and 13 with non-epileptiform abnormalities. This implementation demonstrates that the BC-1 system allows for high-quality EEG acquisition by NEHWs, significantly improving access to neurological diagnostics and treatment.

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